Sawyer Welden

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Experience

Noregon Systems | Software Engineer — Apr 2024 – Present

- Independently built AI-assisted repair-time estimation software with configurable model integrations, .NET APIs, SQL persistence, and human review workflows.
- Modernized data access and expanded regression coverage as part of a six-service .NET migration, preserving existing API contracts.
- Developed C# services for TripVision, a vehicle monitoring platform supporting predictive maintenance across 300,000+ vehicles.
- Designed and implemented billing portal APIs supporting customer management across multiple Noregon products.
- Diagnosed database connection exhaustion with custom load tests and corrected connection lifetimes and transient-failure handling.
- Reduced page-load times by over 40% through SQL stored procedure and Angular optimizations.

Wake Forest School of Medicine | Programmer 3 — Jul 2021 – Dec 2023

- Built isolation-forest anomaly detection for the MoTrPAC clinical trial, migrated a SAS pipeline to R, and developed R Shiny quality-control dashboards.
- Co-developed the aorsf R package for survival analysis, including C++ optimizations to Cox regression.
- Trained Python/scikit-learn models for a posture-aware blood-pressure cuff and presented results to healthcare stakeholders.

Rochester Institute of Technology | Graduate Research Assistant — Jul 2019 – Dec 2021

- Built Julia and React tools for fair-division simulations, survey analysis, and preference modeling.

Technical Skills

Languages: C#, Python, SQL, Java, TypeScript/JavaScript, R, C++
Backend & data: .NET, ASP.NET Core, SQL Server, Spring WebFlux
Cloud & delivery: Azure, Azure DevOps, CI/CD pipelines, Docker, Git
Web & analytics: Angular, R Shiny, scikit-learn

Selected Projects

Piedmont Pauper League & Analytics Tools – *Python, JavaScript* — piedmontpauper.org | sawyerwelden.com/pauper

- Built and maintain a league platform that imports results from North Carolina game stores and updates standings, Elo rankings, and player profiles; created additional tools for deck clustering, turn-by-turn probabilities, and tournament analysis.

Selected Publications

coauthor

Flexible outlier detection in multicenter clinical trials. Journal of Clinical and Translational Science (2026).
Accelerated and Interpretable Oblique Random Survival Forests. Journal of Computational and Graphical Statistics (2024).
aorsf: An R package for supervised learning using the oblique random survival forest. Journal of Open Source Software (2022).

Education

Rochester Institute of Technology – M.S., Computer Science — 2021
New College of Florida – B.A., Computer Science — 2019